Jinliang Liu

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EDUCATION

University of Electronic Science and Technology of China (UESTC) *Sep 2023 – Present*
B.Eng. in Software Engineering (Embedded Systems) Chengdu, China

- **GPA**: 3.8/4.0
- **Relevant Coursework**: Computer Architecture (95), Embedded Operating Systems (94), Artificial Intelligence Era (93), Principles of Database and Application (91), Computer Networks (90).

RESEARCH INTERESTS

Multi-modal LLMs & AI Generative Models: Multi-modal Reasoning, Video, Image, Audio Generation.

Post-Training and RL for Multi-Modal Models: Alignment and Optimization.

AI Agents: Multi-Agent Architectures, Tool Augmentation, Planning and Decomposition of Complex Tasks.

RESEARCH EXPERIENCE

the Chinese University of Hong Kong *Dec 2025 – Present*
Research Intern (Advisor: Dr. Dongchao Yang) — Multi-Modal Language Model Post-Training Hong Kong / Shenzhen

- Investigated post-training RL methods for multimodal generative models, exploring GRPO and DPO combined with variational autoregressive (VAR) practice to better align generative model distributions with human preference.
- Optimized text-to-music and TTS models to reduce hallucinations and improve human preference alignment through multi-objective reward design with compositional scoring across acoustic quality, semantic fidelity, and user preference. Implemented gradient surgery to resolve conflicting gradients across multi-reward objectives.
- Constructed a difficulty-based data filtering curriculum and built the end-to-end post-training infrastructure pipeline covering data preprocessing, training orchestration, and automated evaluation.

Yangtze Delta Region Institute of UESTC *Jul 2025 – Feb 2026*
Research Intern (Advisor: Prof. Shaoning Zeng) — LLMs for RAG and CoT Agent Huzhou, Zhejiang

- Researched RAG frameworks and CoT-based AI agents for LLM reasoning, investigating supervised fine-tuning and reinforcement learning strategies to improve multi-step reasoning capabilities.
- Developed a multi-view knowledge-graph-based RAG framework leveraging attention-head specialization to enhance LLM reasoning on knowledge graphs, achieving SOTA performance on multi-hop question answering benchmarks.

Open Lab of Intelligent Robots, Peking University *Nov 2024 – Jun 2025*
Research Intern (Advisor: Prof. Hao Tang) — Deep Learning-Based Image Segmentation Beijing (Online)

- Applied wavelet transforms for multi-modal vision and EEG feature fusion, capturing complementary spatial and temporal representations across modalities to improve feature richness for segmentation tasks.
- Improved model training stability and segmentation accuracy through deep supervision and multi-level decoder design, validated on surgical and deep-sea exploration datasets.

PUBLICATIONS

[1] **Sole First Author**, "*Lyra: Hierarchical Alignment for Autoregressive Audio Generation via Variational Policy Optimisation*," In preparation, targeting NeurIPS, 2026.

[2] **Sole First Author**, "*Think Parallax: Solving Multi-Hop Problems via Multi-View Knowledge-Graph-Based Retrieval-Augmented Generation*," ACL 2026 Main Conference, arXiv:2510.15552.

[3] **Core Contributor**, "*HeartMuLa: A Family of Open Sourced Music Foundation Models*," Technical Report - Contributed post-training alignment algorithms, applying preference optimization to improve music generation quality., arXiv:2601.10547.

[4] **Third Author**, "*Frequency Domain Enhanced U-Net for Low-Frequency Information-Rich Image Segmentation in Surgical and Deep-Sea Exploration Robots*," Under Review, IJCV, arXiv:2502.03829.

PROJECTS & COMPETITIONS

PPO Training Framework with Advanced Reward Optimization *Jul 2025 – Aug 2025*
Independent Project — PyTorch, Transformers, PPO, GAE

- Implemented a complete PPO-based RLHF pipeline for Qwen3-8B with on-policy sampling, KL-regularized token-level rewards, GAE-based advantage estimation, and actor–critic updates.
- Proposed group-relative reward shaping to mitigate reward-scale variance, reducing critic target drift and stabilizing policy/value convergence.

STM32F4-Based Quadcopter Design *Oct 2024 – Dec 2025*
Project Leader — C, μC/OS-II, Real-time Systems, PCB Design (Advisor: Prof. Yong Liao)

- Engineered the core control and sensing system in C for an autonomous quadcopter, implementing real-time data processing and task scheduling on an embedded platform.

Real-Time Gesture Recognition System *May 2024 – Sep 2024*
Independent Project — PyTorch, ESP-IDF, ONNX Quantization, C++

- Quantized a MobileNet model onto an ESP32-S3 MCU, building a full-stack real-time inference system served via a local server.

CNN for Imbalanced Image Classification *Nov 2023 – Mar 2024*
Independent Project — PyTorch, Computer Vision, Data Augmentation

- Designed a CNN for herbal medicine classification, improving performance on imbalanced data via optimized loss functions and advanced augmentation techniques.

TECHNICAL SKILLS

Languages: Python, C/C++, Shell **Frameworks & Libraries**: PyTorch, TensorFlow, MXNet, LangChain
Developer Tools & Platforms: Git, Linux, ESP-IDF, Altium Designer

HONORS AND PRIZES

Provincial Second Prize, C4 National College Student Network Technology Challenge	Jun 2025
AI & ML Team Lead, Blockchain-Era & Embedded System Innovation Labs, UESTC	Sep 2024 – Present
National First Prize, National College Digital Art & Design Competition (NCDA)	Aug 2024
UESTC Medal — University's Highest Student Honor for well-rounded excellence	May 2024
Model Student Scholarship, Top 5% in Comprehensive Quality Assessment	Sep 2023